José Cotrim

Automation & Software Engineer

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PROFESSIONAL OVERVIEW

.NET Software Engineer with +2 years experience developing production-grade applications using C#, .NET, WPF, Entity Framework and SQL Server for industrial environments. Skilled in software integration, Git, Azure DevOps and CI/CD within Agile environments. Interest in Azure and cloud technologies.

WORK EXPERIENCE

AUTOMATION & SOFTWARE ENGINEER | Preh, Trofa NOV 2025 – PRESENT

- Designed and developed .NET (WPF) applications for configurable production-line parameterization using JSON and SQL Server, eliminating software recompilation for production changes.
- Developed manufacturing traceability applications using .NET (MVVM architecture), SQL Server and stored procedures, ensuring data consistency across automotive production systems.
- Developed .NET WinUI machine vision applications using Entity Framework, MVTec Halcon and SQL Server for automated quality inspection, including defect detection, brightness analysis and symbol verification.
- Designed reusable Halcon vision libraries packaged as NuGet components and deployed as Windows services across production machines.
- Integrated industrial software with Beckhoff TwinCAT 3 PLCs, MQTT, SQL Server and industrial hardware for automotive manufacturing systems.
- Contributed to the migration from ABB Pluto safety controllers to Beckhoff TwinSAFE, including safety I/O definition, electrical schematics and machine validation.
- Integrated and validated complete production stations for Mercedes-Benz manufacturing lines, including machine vision, sensor calibration and quality control.
- Collaborated using Azure DevOps, Git and CI/CD pipelines within an Agile software development environment.

PRODUCT DEVELOPMENT ENGINEER | INEGI, Location SEP 2024 – OCT 2025

- Developed LabVIEW applications for high-speed experimental testing systems.
- Built Python tools to automate questionnaire processing and statistical analysis.
- Developed AI-based body tracking algorithms for applications using ZED 2i for posture analysis, body tracking and object detection.

EDUCATION

Masters in Automation| Faculty of Engineering of University of Porto (FEUP), Porto Graduated: 2024
Bachelor of Mechanical Engineering | Faculty of Engineering of University of Porto (FEUP), Porto Graduated: 2022

SKILLS

C#, .NET, WPF, SQL Server, REST APIs, MVVM, Entity Framework, Azure DevOps, Python, TwinCAT 3, Codesys, ST/FBD/Ladder, LabVIEW, Siemens TIA Portal, WinForms, MQTT

LANGUAGES

Portuguese Native, **English** C1, **German** A2